Data Analysis
AMS315

Chapter 10

Two-Sample Tests

Learning Objectives

In this chapter, you learn:

- How to use hypothesis testing for comparing the difference between
  - The means of two independent populations
  - The means of two related populations
  - The proportions of two independent populations
  - The variances of two independent populations

Two-Sample Tests

Two-Sample Tests

Population Means, Independent Samples	Populations Means, Related Samples	Population Proportions	Population Variances

Examples:

Population 1 vs. independent Population 2	Same population before vs. after treatment	Proportion 1 vs. Proportion 2	Variance 1 vs. Variance 2

Difference Between Two Means

Population means, independent samples *

- σ_1 and σ_2 known
- σ_1 and σ_2 unknown, assumed equal
- σ_1 and σ_2 unknown, not assumed equal

Goal: Test hypothesis or form a confidence interval for the difference between two population means, $\mu_1 - \mu_2$

The point estimate for the difference is

$$\bar{X}_1 - \bar{X}_2$$

Independent Samples

Population means, independent samples *

- σ_1 and σ_2 known
- σ_1 and σ_2 unknown, assumed equal
- σ_1 and σ_2 unknown, not assumed equal

- Different data sources
  - Unrelated
  - Independent
    - Sample selected from one population has no effect on the sample selected from the other population
- Use the difference between 2 sample means
- Use Z test, a pooled-variance t test, or a separate-variance t test

Difference Between Two Means

Population means, independent samples *

- σ_1 and σ_2 known → Use a **Z** test statistic
- σ_1 and σ_2 unknown, assumed equal → Use S_p to estimate unknown σ , use a **t** test statistic and pooled standard deviation
- σ_1 and σ_2 unknown, not assumed equal → Use S_1 and S_2 to estimate unknown σ_1 and σ_2, use a separate-variance **t** test

σ_1 and σ_2 Known

Population means, independent samples

- σ_1 and σ_2 known *
- σ_1 and σ_2 unknown, assumed equal
- σ_1 and σ_2 unknown, not assumed equal

Assumptions:

- Samples are randomly and independently drawn
- Population distributions are normal or both sample sizes are ≥ 30
- Population standard deviations are known

σ_1 and σ_2 Known

(continued)

Population means, independent samples

- σ_1 and σ_2 known *
- σ_1 and σ_2 unknown, assumed equal
- σ_1 and σ_2 unknown, not assumed equal

When σ_1 and σ_2 are known and both populations are normal or both sample sizes are at least 30, the test statistic is a Z-value…

…and the standard error of $\bar{X}_1 - \bar{X}_2$ is

$$\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

σ_1 and σ_2 Known

(continued)

Population means, independent samples

- σ_1 and σ_2 known *
- σ_1 and σ_2 unknown, assumed equal
- σ_1 and σ_2 unknown, not assumed equal

The test statistic for $\mu_1 - \mu_2$ is:

$$Z = \frac{\left(\bar{X}_1 - \bar{X}_2\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Hypothesis Tests for Two Population Means

Two Population Means, Independent Samples

Lower-tail test:	Upper-tail test:	Two-tail test:
$H_0: \mu_1 \geq \mu_2$ $H_1: \mu_1 < \mu_2$	$H_0: \mu_1 \leq \mu_2$ $H_1: \mu_1 > \mu_2$	$H_0: \mu_1 = \mu_2$ $H_1: \mu_1 \neq \mu_2$
i.e.,	i.e.,	i.e.,
$H_0: \mu_1 - \mu_2 \geq 0$ $H_1: \mu_1 - \mu_2 < 0$	$H_0: \mu_1 - \mu_2 \leq 0$ $H_1: \mu_1 - \mu_2 > 0$	$H_0: \mu_1 - \mu_2 = 0$ $H_1: \mu_1 - \mu_2 \neq 0$

Hypothesis tests for $\mu_1 - \mu_2$

Two Population Means, Independent Samples

Lower-tail test:

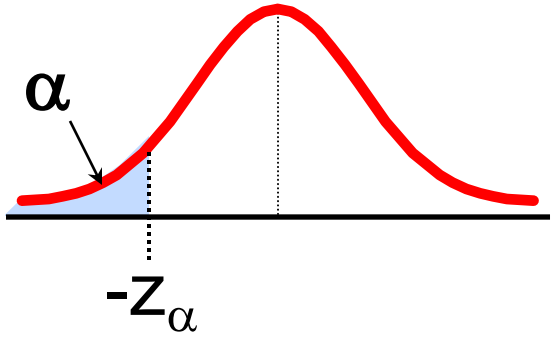
$H_0: \mu_1 - \mu_2 \geq 0$
$H_1: \mu_1 - \mu_2 < 0$

Upper-tail test:

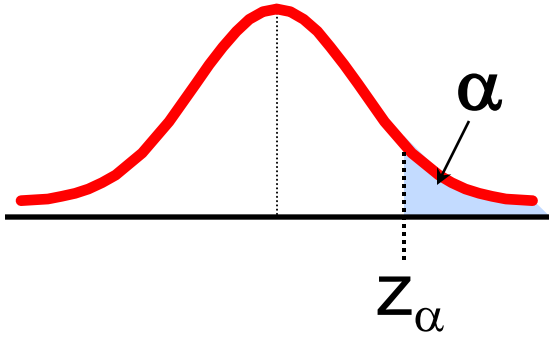
$H_0: \mu_1 - \mu_2 \leq 0$
$H_1: \mu_1 - \mu_2 > 0$

Two-tail test:

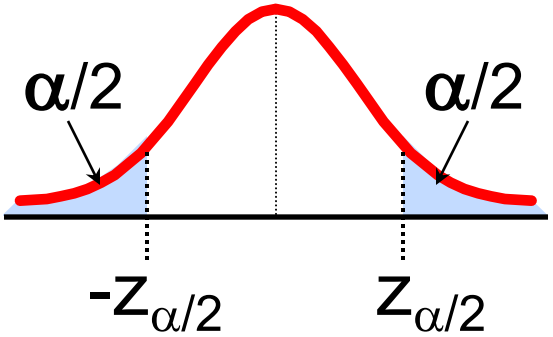
$H_0: \mu_1 - \mu_2 = 0$
$H_1: \mu_1 - \mu_2 \neq 0$



Reject H_0 if $Z < -Z_\alpha$



Reject H_0 if $Z > Z_\alpha$



Reject H_0 if $Z < -Z_{\alpha/2}$ or $Z > Z_{\alpha/2}$

Confidence Interval, σ_1 and σ_2 Known

Population means, independent samples

- σ_1 and σ_2 known *
- σ_1 and σ_2 unknown, assumed equal
- σ_1 and σ_2 unknown, not assumed equal

The confidence interval for $\mu_1 - \mu_2$ is:

$$\left(\bar{X}_1 - \bar{X}_2\right) \pm Z\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

σ_1 and σ_2 Unknown, Assumed Equal

Population means, independent samples

- σ_1 and σ_2 known
- σ_1 and σ_2 unknown, assumed equal *
- σ_1 and σ_2 unknown, not assumed equal

Assumptions:

- Samples are randomly and independently drawn
- Populations are normally distributed or both sample sizes are at least 30
- Population variances are unknown but assumed equal

σ_1 and σ_2 Unknown, Assumed Equal

(continued)

Population means, independent samples

- σ_1 and σ_2 known
- σ_1 and σ_2 unknown, assumed equal *
- σ_1 and σ_2 unknown, not assumed equal

Forming interval estimates:

- The population variances are assumed equal, so use the two sample variances and pool them to estimate the common σ^2
- the test statistic is a t value with $(n_1 + n_2 - 2)$ degrees of freedom

σ_1 and σ_2 Unknown, Assumed Equal

(continued)

Population means, independent samples

- σ_1 and σ_2 known
- σ_1 and σ_2 unknown, assumed equal *
- σ_1 and σ_2 unknown, not assumed equal

The pooled variance is

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 - 1) + (n_2 - 1)}$$

σ_1 and σ_2 Unknown, Assumed Equal

(continued)

Population means, independent samples

- σ_1 and σ_2 known
- σ_1 and σ_2 unknown, assumed equal *
- σ_1 and σ_2 unknown, not assumed equal

The test statistic for $\mu_1 - \mu_2$ is:

$$t = \frac{\left(\bar{X}_1 - \bar{X}_2\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

Where t has $(n_1 + n_2 - 2)$ d.f.,

and

$$S_p^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 - 1) + (n_2 - 1)}$$

Confidence Interval, σ_1 and σ_2 Unknown

Population means, independent samples

- σ_1 and σ_2 known
- σ_1 and σ_2 unknown, assumed equal *
- σ_1 and σ_2 unknown, not assumed equal

The confidence interval for $\mu_1 - \mu_2$ is:

$$\left(\bar{X}_1 - \bar{X}_2\right) \pm t_{n_1+n_2-2}\sqrt{S_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$$

Where

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{(n_1-1)+(n_2-1)}$$

Pooled-Variance t Test: Example

You are a financial analyst for a brokerage firm. Is there a difference in dividend yield between stocks listed on the NYSE & NASDAQ? You collect the following data:

	NYSE	NASDAQ
Number	**21**	**25**
Sample mean	**3.27**	**2.53**
Sample std dev	**1.30**	**1.16**

Assuming both populations are approximately normal with equal variances, is there a difference in average yield (α = 0.05)?

Calculating the Test Statistic

The test statistic is:

$$t = \frac{\left(\bar{X}_1 - \bar{X}_2\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{\left(3.27 - 2.53\right) - 0}{\sqrt{1.5021\left(\frac{1}{21} + \frac{1}{25}\right)}} = \boxed{2.040}$$

$$S_p^2 = \frac{(n_1 - 1)S_1^{\ 2} + (n_2 - 1)S_2^{\ 2}}{(n_1 - 1) + (n_2 - 1)} = \frac{(21 - 1)1.30^2 + (25 - 1)1.16^2}{(21\text{-}1) + (25 - 1)} = 1.5021$$

Solution

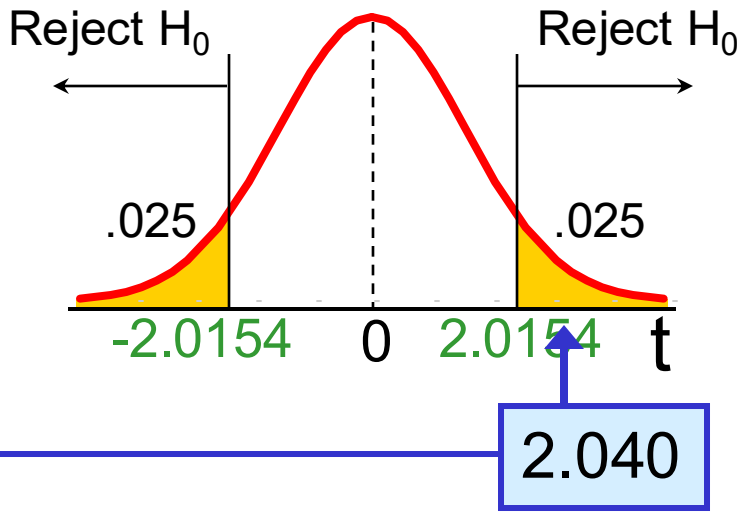
H_0: $\mu_1 - \mu_2 = 0$ i.e. $(\mu_1 = \mu_2)$

H_1: $\mu_1 - \mu_2 \neq 0$ i.e. $(\mu_1 \neq \mu_2)$

$\alpha = 0.05$

df = 21 + 25 - 2 = 44

Critical Values: t = ± 2.0154



Test Statistic:

$$t = \frac{3.27 - 2.53}{\sqrt{1.5021\left(\frac{1}{21} + \frac{1}{25}\right)}} = \boxed{2.040}$$

Decision:

Reject H_0 at $\alpha = 0.05$

Conclusion:

There is evidence of a difference in means.

σ_1 and σ_2 Unknown, Not Assumed Equal

Population means, independent samples

- σ_1 and σ_2 known
- σ_1 and σ_2 unknown, assumed equal
- σ_1 and σ_2 unknown, not assumed equal *

Assumptions:

- Samples are randomly and independently drawn
- Populations are normally distributed or both sample sizes are at least 30
- Population variances are unknown but cannot be assumed to be equal

σ_1 and σ_2 Unknown, Not Assumed Equal

(continued)

Population means, independent samples

- σ_1 and σ_2 known
- σ_1 and σ_2 unknown, assumed equal
- σ_1 and σ_2 unknown, not assumed equal *

Forming the test statistic:

- The population variances are not assumed equal, so include the two sample variances in the computation of the t-test statistic
- the test statistic is a t value with *v* degrees of freedom (see next slide)

σ_1 and σ_2 Unknown, Not Assumed Equal

(continued)

Population means, independent samples

- σ_1 and σ_2 known
- σ_1 and σ_2 unknown, assumed equal
- σ_1 and σ_2 unknown, not assumed equal *

The number of degrees of freedom is the integer portion of:

$$\nu = \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{\left(\frac{S_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{S_2^2}{n_2}\right)^2}{n_2 - 1}}$$

σ_1 and σ_2 Unknown, Not Assumed Equal

(continued)

Population means, independent samples

- σ_1 and σ_2 known
- σ_1 and σ_2 unknown, assumed equal
- σ_1 and σ_2 unknown, not assumed equal *

The test statistic for $\mu_1 - \mu_2$ is:

$$t = \frac{\left(\bar{X}_1 - \bar{X}_2\right) - \left(\mu_1 - \mu_2\right)}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}}$$

Related Populations

Related samples

Tests Means of 2 Related Populations

- Paired or matched samples
- Repeated measures (before/after)
- Use difference between paired values:

$$D_i = X_{1i} - X_{2i}$$

- Eliminates Variation Among Subjects
- Assumptions:
  - Both Populations Are Normally Distributed
  - Or, if not Normal, use large samples

Mean Difference, σ_D Known

Related samples

The ith paired difference is D_i , where

$$D_i = X_{1i} - X_{2i}$$

The point estimate for the population mean paired difference is $\bar{D}$:

$$\bar{D} = \frac{\sum_{i=1}^{n} D_i}{n}$$

Suppose the population standard deviation of the difference scores, σ_D, is known

n is the number of pairs in the paired sample

Mean Difference, σ_D Known

(continued)

Paired samples

The test statistic for the mean difference is a Z value:

$$Z = \frac{\bar{D} - \mu_D}{\frac{\sigma_D}{\sqrt{n}}}$$

Where

μ_D = hypothesized mean difference

σ_D = population standard dev. of differences

n = the sample size (number of pairs)

Confidence Interval, σ_D Known

Paired samples

The confidence interval for μ_D is

$$\bar{D} \pm Z \frac{\sigma_D}{\sqrt{n}}$$

Where

n = the sample size

(number of pairs in the paired sample)

Mean Difference, σ_D Unknown

Related samples

If σ_D is unknown, we can estimate the unknown population standard deviation with a sample standard deviation:

The sample standard deviation is

$$S_D = \sqrt{\frac{\sum_{i=1}^{n}(D_i - \bar{D})^2}{n-1}}$$

Mean Difference, σ_D Unknown

(continued)

Paired samples

- Use a paired t test, the test statistic for $\bar{D}$ is now a t statistic, with n-1 d.f.:

$$t = \frac{\bar{D} - \mu_D}{\frac{S_D}{\sqrt{n}}}$$

Where t has n - 1 d.f.

and S_D is: $S_D = \sqrt{\frac{\sum_{i=1}^{n}(D_i - \bar{D})^2}{n-1}}$

Confidence Interval, σ_D Unknown

Paired samples

The confidence interval for μ_D is

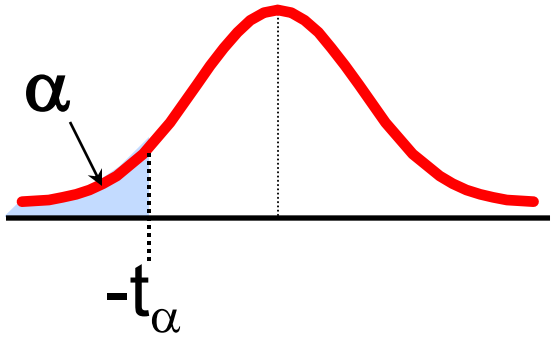
$$\bar{D} \pm t_{n-1} \frac{S_D}{\sqrt{n}}$$

where $$S_D = \sqrt{\frac{\sum_{i=1}^{n} (D_i - \bar{D})^2}{n-1}}$$

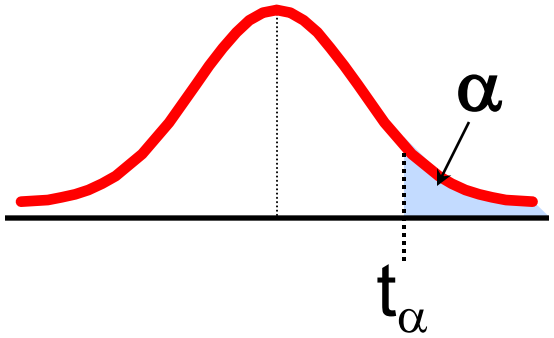
Hypothesis Testing for Mean Difference, σ_D Unknown

Paired Samples

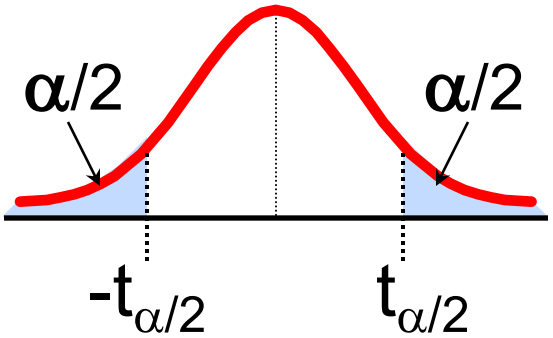
Lower-tail test:	Upper-tail test:	Two-tail test:
$H_0: \mu_D \geq 0$ $H_1: \mu_D < 0$	$H_0: \mu_D \leq 0$ $H_1: \mu_D > 0$	$H_0: \mu_D = 0$ $H_1: \mu_D \neq 0$



Reject H_0 if $t < -t_\alpha$



Reject H_0 if $t > t_\alpha$



Reject H_0 if $t < -t_{\alpha/2}$ or $t > t_{\alpha/2}$

Where t has $n - 1$ d.f.

Paired t Test Example

- Assume you send your salespeople to a "customer service" training workshop. Has the training made a difference in the number of complaints? You collect the following data:

Salesperson	Number of Complaints: Before (1)	After (2)	(2) - (1) Difference, D_i
C.B.	6	4	- 2
T.F.	20	6	-14
M.H.	3	2	- 1
R.K.	0	0	0
M.O.	4	0	- 4
			-21

$$\bar{D} = \frac{\sum D_i}{n}$$

$$= -4.2$$

$$S_D = \sqrt{\frac{\sum (D_i - \bar{D})^2}{n-1}}$$

$$= 5.67$$

Paired t Test: Solution

- Has the training made a difference in the number of complaints (at the 0.01 level)?

$$H_0: \mu_D = 0$$
$$H_1: \mu_D \neq 0$$

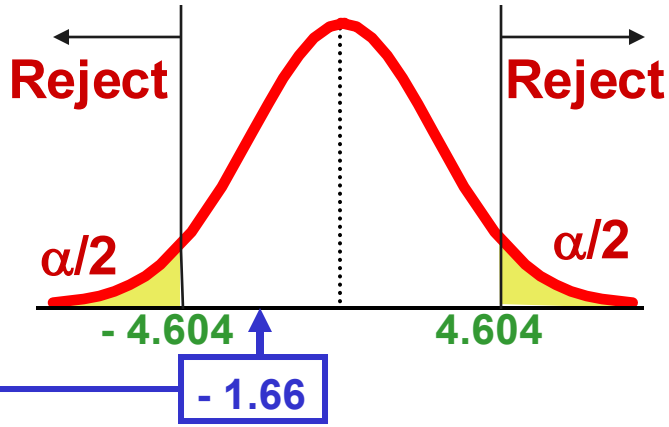
$\alpha = .01$ $\bar{D} = -4.2$

Critical Value = ± 4.604

d.f. = n - 1 = 4

Test Statistic:

$$t = \frac{\bar{D} - \mu_D}{S_D/\sqrt{n}} = \frac{-4.2 - 0}{5.67/\sqrt{5}} = \boxed{-1.66}$$



Decision: Do not reject H_0

(t stat is not in the reject region)

Conclusion: There is not a significant change in the number of complaints.

Two Population Proportions

Population proportions

Goal: test a hypothesis or form a confidence interval for the difference between two population proportions,

$$\pi_1 - \pi_2$$

Assumptions:

$n_1 \pi_1 \geq 5$, $n_1(1- \pi_1) \geq 5$

$n_2 \pi_2 \geq 5$, $n_2(1- \pi_2) \geq 5$

The point estimate for the difference is $p_1 - p_2$

Two Population Proportions

Population proportions

Since we begin by assuming the null hypothesis is true, we assume $\pi_1 = \pi_2$ and pool the two sample estimates

The pooled estimate for the overall proportion is:

$$\bar{p} = \frac{X_1 + X_2}{n_1 + n_2}$$

where X_1 and X_2 are the numbers from samples 1 and 2 with the characteristic of interest

Two Population Proportions

(continued)

Population proportions

The test statistic for

$p_1 - p_2$ is a Z statistic:

$$Z = \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

where $\bar{p} = \frac{X_1 + X_2}{n_1 + n_2}$, $p_1 = \frac{X_1}{n_1}$, $p_2 = \frac{X_2}{n_2}$

Confidence Interval for Two Population Proportions

Population proportions

The confidence interval for $\pi_1 - \pi_2$ is:

$$(p_1 - p_2) \pm Z\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

Hypothesis Tests for Two Population Proportions

Population proportions

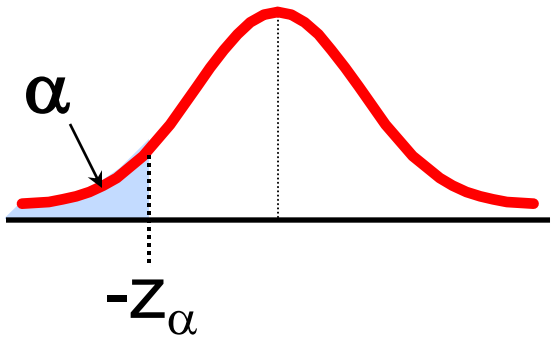
Lower-tail test:	Upper-tail test:	Two-tail test:
$H_0: \pi_1 \geq \pi_2$ $H_1: \pi_1 < \pi_2$	$H_0: \pi_1 \leq \pi_2$ $H_1: \pi_1 > \pi_2$	$H_0: \pi_1 = \pi_2$ $H_1: \pi_1 \neq \pi_2$
i.e.,	i.e.,	i.e.,
$H_0: \pi_1 - \pi_2 \geq 0$ $H_1: \pi_1 - \pi_2 < 0$	$H_0: \pi_1 - \pi_2 \leq 0$ $H_1: \pi_1 - \pi_2 > 0$	$H_0: \pi_1 - \pi_2 = 0$ $H_1: \pi_1 - \pi_2 \neq 0$

Hypothesis Tests for Two Population Proportions

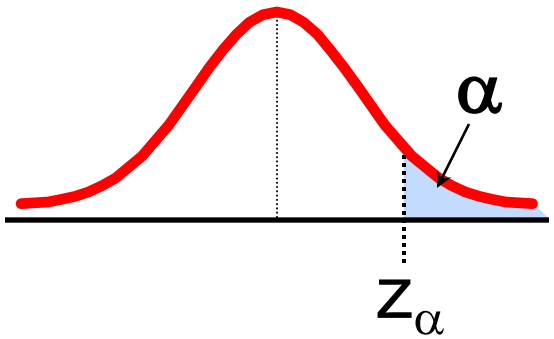
(continued)

Population proportions

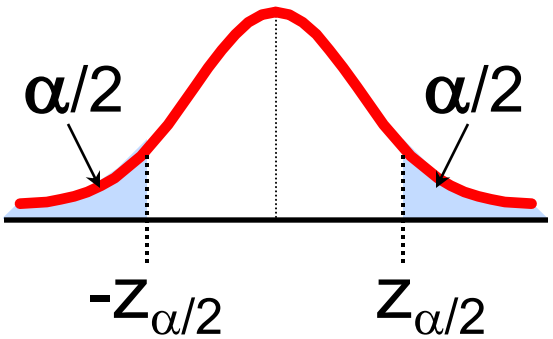
Lower-tail test:	Upper-tail test:	Two-tail test:
$H_0: \pi_1 - \pi_2 \geq 0$	$H_0: \pi_1 - \pi_2 \leq 0$	$H_0: \pi_1 - \pi_2 = 0$
$H_1: \pi_1 - \pi_2 < 0$	$H_1: \pi_1 - \pi_2 > 0$	$H_1: \pi_1 - \pi_2 \neq 0$



Reject H_0 if $Z < -Z_\alpha$



Reject H_0 if $Z > Z_\alpha$



Reject H_0 if $Z < -Z_{\alpha/2}$ or $Z > Z_{\alpha/2}$

Example: Two population Proportions

Is there a significant difference between the proportion of men and the proportion of women who will vote Yes on Proposition A?

- In a random sample, 36 of 72 men and 31 of 50 women indicated they would vote Yes
- Test at the .05 level of significance

Example: Two population Proportions

(continued)

- The hypothesis test is:

H_0: $\pi_1 - \pi_2 = 0$ (the two proportions are equal)

H_1: $\pi_1 - \pi_2 \neq 0$ (there is a significant difference between proportions)

- The sample proportions are:

  - Men: $p_1 = 36/72 = .50$
  - Women: $p_2 = 31/50 = .62$

- The pooled estimate for the overall proportion is:

$$\bar{p} = \frac{X_1 + X_2}{n_1 + n_2} = \frac{36 + 31}{72 + 50} = \frac{67}{122} = .549$$

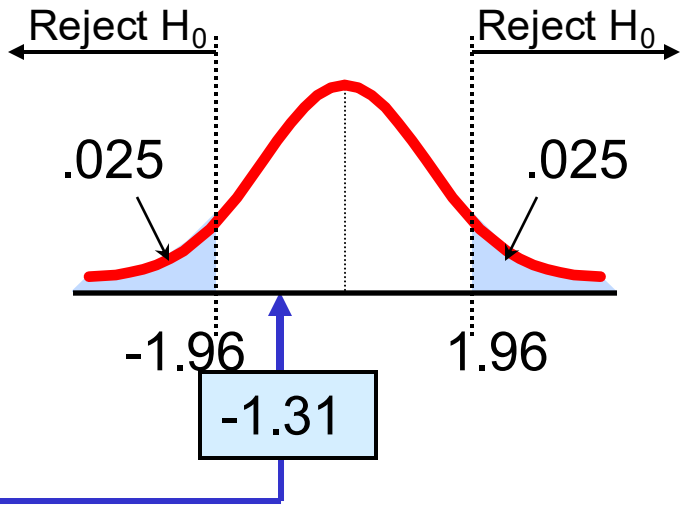
Example: Two population Proportions

(continued)

The test statistic for $\pi_1 - \pi_2$ is:

$$z = \frac{(p_1 - p_2) - (\pi_1 - \pi_2)}{\sqrt{\bar{p}(1-\bar{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$= \frac{(.50 - .62) - (0)}{\sqrt{.549(1-.549)\left(\frac{1}{72} + \frac{1}{50}\right)}} = \boxed{-1.31}$$



Critical Values = ±1.96
For α = .05

Decision: Do not reject H_0

Conclusion: There is not significant evidence of a difference in proportions who will vote yes between men and women.

Hypothesis Tests for Variances

Tests for Two Population Variances *

F test statistic

$H_0: \sigma_1^2 = \sigma_2^2$
$H_1: \sigma_1^2 \neq \sigma_2^2$

Two-tail test

$H_0: \sigma_1^2 \geq \sigma_2^2$
$H_1: \sigma_1^2 < \sigma_2^2$

Lower-tail test

$H_0: \sigma_1^2 \leq \sigma_2^2$
$H_1: \sigma_1^2 > \sigma_2^2$

Upper-tail test

Hypothesis Tests for Variances

(continued)

Tests for Two Population Variances

F test statistic *

The F test statistic is:

$$F = \frac{S_1^2}{S_2^2}$$

S_1^2 = Variance of Sample 1

$n_1 - 1$ = numerator degrees of freedom

S_2^2 = Variance of Sample 2

$n_2 - 1$ = denominator degrees of freedom

The F Distribution

- The F critical value is found from the F table
- There are two appropriate degrees of freedom: numerator and denominator

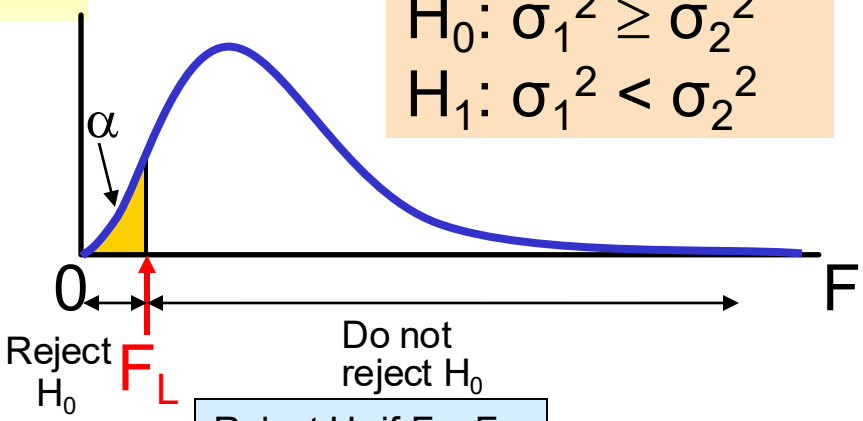
$$F = \frac{S_1^2}{S_2^2}$$

where $df_1 = n_1 - 1$; $df_2 = n_2 - 1$

- In the F table,
  - numerator degrees of freedom determine the column
  - denominator degrees of freedom determine the row

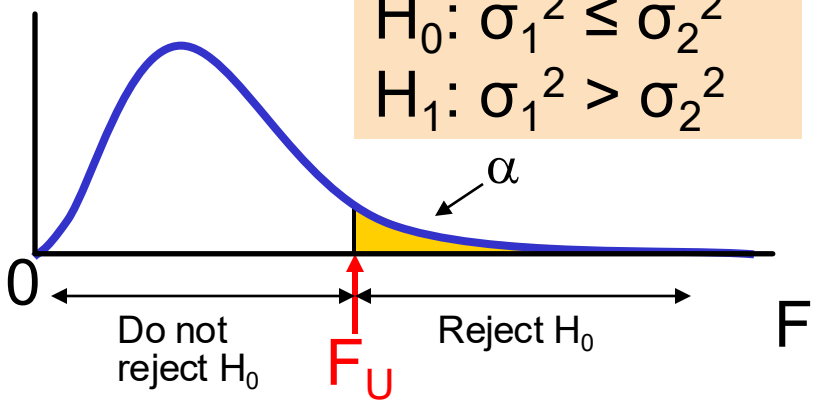
Finding the Rejection Region

$H_0: \sigma_1^2 \geq \sigma_2^2$
$H_1: \sigma_1^2 < \sigma_2^2$



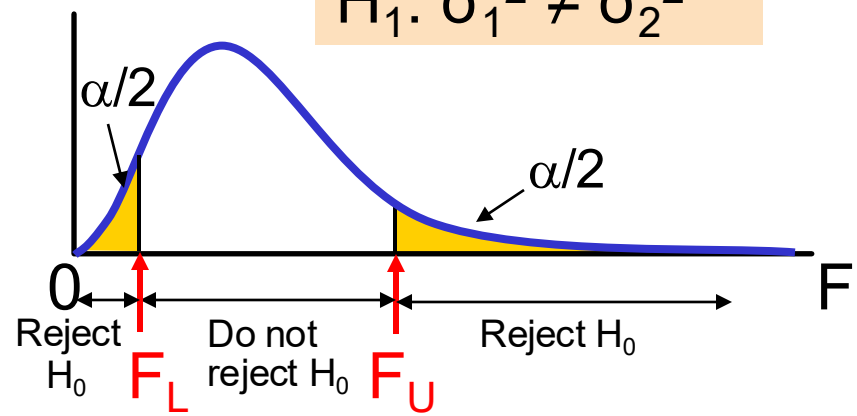
Reject H_0 if $F < F_L$

$H_0: \sigma_1^2 \leq \sigma_2^2$
$H_1: \sigma_1^2 > \sigma_2^2$



Reject H_0 if $F > F_U$

$H_0: \sigma_1^2 = \sigma_2^2$
$H_1: \sigma_1^2 \neq \sigma_2^2$



- rejection region for a two-tail test is:

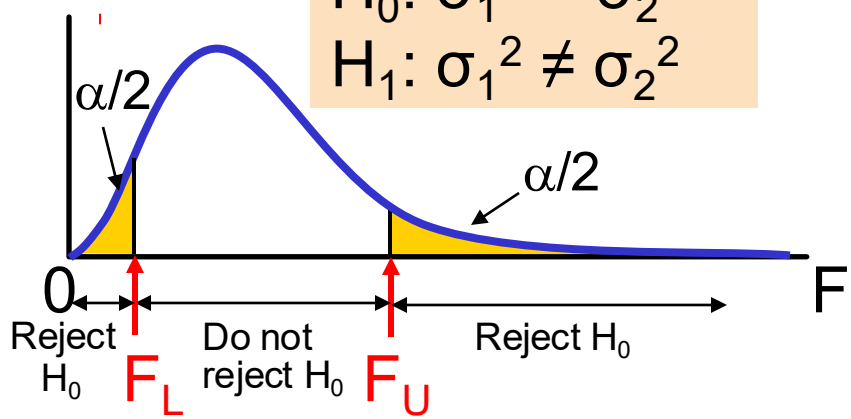
$$F = \frac{S_1^2}{S_2^2} > F_U$$

$$F = \frac{S_1^2}{S_2^2} < F_L$$

Finding the Rejection Region

(continued)

$H_0: \sigma_1^2 = \sigma_2^2$
$H_1: \sigma_1^2 \neq \sigma_2^2$



To find the critical F values:

1. Find F_U from the F table for $n_1 - 1$ numerator and $n_2 - 1$ denominator degrees of freedom

2. Find F_L using the formula:

$$F_L = \frac{1}{F_{U^*}}$$

Where F_{U^*} is from the F table with $n_2 - 1$ numerator and $n_1 - 1$ denominator degrees of freedom (i.e., switch the d.f. from F_U)

F Test: An Example

You are a financial analyst for a brokerage firm. You want to compare dividend yields between stocks listed on the NYSE & NASDAQ. You collect the following data:

	NYSE	NASDAQ
Number	**21**	**25**
Mean	**3.27**	**2.53**
Std dev	**1.30**	**1.16**

Is there a difference in the variances between the NYSE & NASDAQ at the $\alpha = 0.05$ level?

F Test: Example Solution

- Form the hypothesis test:

$H_0: \sigma^2_1 - \sigma^2_2 = 0$ (there is no difference between variances)

$H_1: \sigma^2_1 - \sigma^2_2 \neq 0$ (there is a difference between variances)

- Find the F critical values for $\alpha = 0.05$:

F_U:

- Numerator:
  - $n_1 - 1 = 21 - 1 = 20$ d.f.
- Denominator:
  - $n_2 - 1 = 25 - 1 = 24$ d.f.

$$F_U = F_{.025,\,20,\,24} = 2.33$$

F_L:

- Numerator:
  - $n_2 - 1 = 25 - 1 = 24$ d.f.
- Denominator:
  - $n_1 - 1 = 21 - 1 = 20$ d.f.

$$F_L = 1/F_{.025,\,24,\,20} = 1/2.41 = 0.415$$

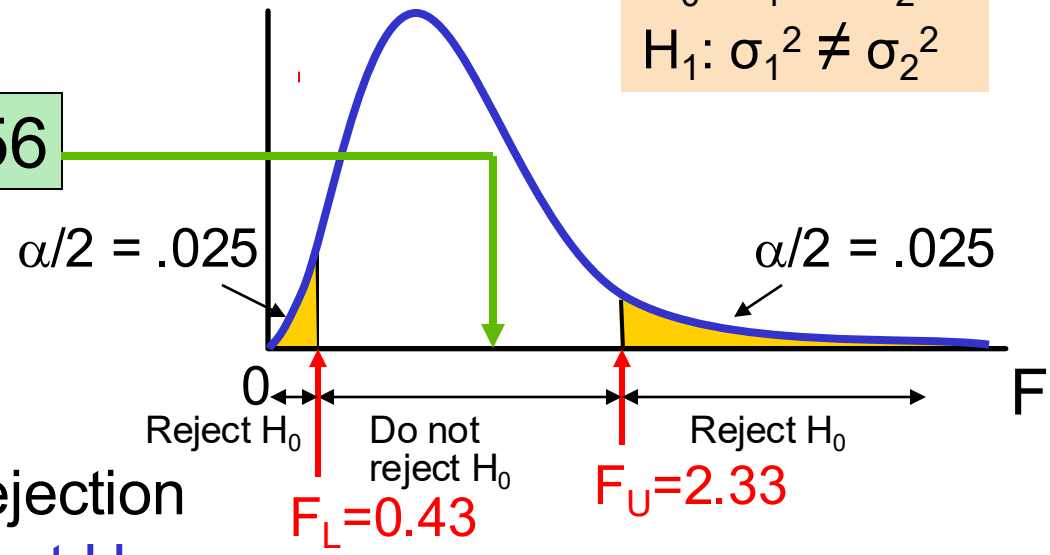
F Test: Example Solution

(continued)

- The test statistic is:

$$F = \frac{S_1^2}{S_2^2} = \frac{1.30^2}{1.16^2} = \boxed{1.256}$$

H_0: $\sigma_1^2 = \sigma_2^2$
H_1: $\sigma_1^2 \neq \sigma_2^2$



- F = 1.256 is not in the rejection region, so we do not reject H_0

- Conclusion: There is not sufficient evidence of a difference in variances at α = .05

F Test: Class Example 1

A researcher is studying the variability in daily exercise times between two groups of people. Group X consists of people who work from home, and Group Y consists of people who work in an office. The researcher wants to know if there is a significant difference in the variances of exercise times between these two groups at a 5% significance level. Assume exercise times are normally distributed.

- From Group X, a random sample of $n_X = 16$ people has a sample variance of $s_X^2 = 40$.
- From Group Y, a random sample of $n_Y = 11$ people has a sample variance of $s_Y^2 = 15$.

Conduct an F-test to determine if there is a significant difference in variances between the two groups.

F Test: Class Example 2

Two brands of tires, Brand A and Brand B, are being compared for variability in their durability. A random sample of 13 tires from Brand A and 16 tires from Brand B are tested for durability, and the sample variances are calculated. The results are as follows:

- Brand A: sample variance $s_A^2 = 36$
- Brand B: sample variance $s_B^2 = 16$

You want to test if there is a significant difference in the variances of tire durability between the two brands at a 1% significance level using an appropriate test.

F Test: Class Example 3

A researcher is studying the reaction times of two different groups of participants, Group X and Group Y, under different conditions. The researcher suspects that there might be a difference in the variability of reaction times between the two groups. A sample of 21 participants from Group X has a variance of $s_X^2 = 50$, and a sample of 16 participants from Group Y has a variance of $s_Y^2 = 30$. Test if there is a significant difference in the variances of reaction times between the two groups at a 5% significance level using an appropriate test.

Two-Sample Tests in EXCEL

For independent samples:

- Independent sample Z test with variances known:
  - Tools | data analysis | z-test: two sample for means
- Pooled variance t test:
  - Tools | data analysis | t-test: two sample assuming equal variances
- Separate-variance t test:
  - Tools | data analysis | t-test: two sample assuming unequal variances

For paired samples (t test):

- Tools | data analysis | t-test: paired two sample for means

For variances:

- F test for two variances:
  - Tools | data analysis | F-test: two sample for variances

Chapter Summary

- Compared two independent samples
  - Performed Z test for the difference in two means
  - Performed pooled variance t test for the difference in two means
  - Performed separate-variance t test for difference in two means
  - Formed confidence intervals for the difference between two means
- Compared two related samples (paired samples)
  - Performed paired sample Z and t tests for the mean difference
  - Formed confidence intervals for the mean difference

Chapter Summary

(continued)

- Compared two population proportions
  - Formed confidence intervals for the difference between two population proportions
  - Performed Z-test for two population proportions
- Performed F tests for the difference between two population variances
- Used the F table to find F critical values